

EXPLORATORY DATA ANALYSIS



Analyzing Spotify Dataset to discover hidden trends and classify song as Hit or Not Hit

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INTRODUCTION

Background:

Spotify is one of the world's most influential music streaming platforms, offering access to millions of songs across diverse genres, artists, and cultures. The central aim of this project is to **investigate what defines a "hit song" on Spotify** by exploring the relationships among these measurable attributes. By analyzing the dataset, this study seeks to identify the patterns, trends, and correlations that distinguish high-performing tracks from less popular ones.

Relevance:

For **Spotify's recommendation system**, insights from this analysis can contribute to improved music discovery and personalized listening experiences. For **artists and producers**, understanding these patterns can help refine creative decisions and production choices that align with listener preferences. Similarly, **marketing teams and record labels** can leverage these findings to predict and promote potential hit tracks more effectively.

Motivation:

The **motivation** behind this work lies in bridging creativity and analytics — transforming musical intuition into quantifiable insights. By examining Spotify's dataset through a data science lens, this project aims to translate subjective musical appeal into objective, interpretable metrics. The analysis not only helps define what makes a track successful but also demonstrates how data-driven exploration can reveal deeper cultural and artistic trends in the modern music industry.

In summary, this project provides a structured exploration of Spotify data to identify the factors that make certain songs stand out. By combining descriptive statistics, visual analytics, and feature-based evaluation, the study aims to highlight the measurable elements that define a hit song and their broader implications for the streaming ecosystem.

OBJECTIVES

The main objective of this project is to analyze Spotify's dataset to uncover the measurable factors that distinguish hit songs from non-hits. The analysis focuses on cleaning, exploring, testing, and modeling data to identify patterns that explain song popularity.

1. Data Cleaning and Preprocessing

The first step is to ensure the dataset is accurate, consistent, and ready for analysis. This involves

- Identifying and handling missing or duplicate records
- Correcting data inconsistencies
- Standardizing feature formats.

Proper preprocessing ensures the quality and reliability of subsequent analyses, allowing for more accurate insights and model performance.

2. Feature Exploration and Correlation Analysis

This objective focuses on understanding how individual musical attributes relate to a song's popularity. By calculating correlations and examining feature distributions, we aim to identify which musical characteristics strongly influence audience appeal. This step helps to establish foundational relationship that may explain why some tracks gain more traction than others.

3. Statistical Testing

To move beyond observation and validate findings, statistical methods like *t-tests* and *ANOVA* are applied to evaluate hypotheses about feature differences between hit and non-hit songs. These tests determine whether variations in features such as energy or danceability are statistically significant or occur by chance. This objective adds scientific rigor to the analysis by supporting or rejecting hypotheses with quantifiable evidence.

4. Feature Engineering

This phase involves transforming and creating new variables that may improve model interpretability and predictive capability. For example, continuous features can be normalized or categorized, and derived metrics (like duration in minutes or energy-density ratios) can be generated. Well-engineered features can enhance model performance and reveal deeper relationships between musical attributes and popularity.

5. Visualization and Interpretation

Visual analytics play a key role in uncovering hidden patterns and making results more interpretable.

Various graphs and plot that can be used are:

- Histograms
- Scatter plots
- Heatmaps
- Boxplots

This objective aims to visualize how key features influence popularity levels. Visualizations help communicate findings clearly, enabling understanding of the data's underlying trends and correlations.

6. Predictive Modeling

The final objective is to apply machine learning techniques to classify songs as **hit** or **non-hit** based on their features.

A basic predictive model such as **Logistic Regression**, **Decision Tree** or **RandomForrestClassifer** is developed, trained, and evaluated using accuracy and other performance metrics. This step not only evaluates the practical applicability of the findings but also demonstrates how data analytics can be used to predict real-world outcomes in the music industry.

Together, these objectives establish a structured framework for understanding the factors that contribute to a song's success on Spotify. By progressing systematically from data cleaning and feature exploration to statistical validation, visualization, and predictive modeling, the study ensures both analytical rigor and practical relevance. Achieving these goals will provide meaningful insights into how musical and technical features influence popularity, bridging the gap between artistic creativity and data-driven decision-making in the modern music industry.

PROBLEM STATEMENT

In today's digital music landscape, platforms like **Spotify** generate massive amounts of data on songs, artists, and listener behavior. Each track is described by quantifiable attributes such as **tempo, energy, danceability, loudness, and valence**, which reflect its musical composition and production quality. However, despite this abundance of data, identifying what truly defines a "hit song" remains a complex challenge. Popularity is influenced by a combination of technical, musical, and cultural factors — many of which are not yet fully understood from a data-driven perspective.

Traditional methods of predicting song success have often relied on **subjective judgments**, intuition, or post-release audience reactions. These approaches lack analytical depth and fail to leverage the rich quantitative features available in modern music databases. Moreover, while Spotify's recommendation algorithms excel at suggesting similar tracks to listeners, they are not explicitly designed to **analyze or explain** the underlying traits that make certain songs achieve higher popularity. This creates a gap between **music recommendation** and **music understanding**.

The project therefore aims to **bridge this gap** by applying data analytics and statistical techniques to uncover measurable factors that distinguish *hit songs* from *non-hits*. It focuses on cleaning and standardizing Spotify data to ensure consistency, exploring correlations among musical and cultural attributes, and testing statistical assumptions to validate feature significance.

By addressing these issues, the project seeks to answer a fundamental question:

"What measurable musical and technical factors define a hit song on Spotify, and how can data-driven insights help explain or predict a track's popularity?"

Through this analysis, the study not only identifies the characteristics that contribute to a song's success but also provides a data-backed foundation for enhancing recommendation systems, guiding artist strategies, and informing marketing decisions within the music industry.

LITERATURE REVIEW

Research on predicting song popularity and analyzing Spotify audio features has matured into two complementary threads:

- Feature-driven predictive models that combine audio, metadata, and sometimes social signals
- Descriptive studies that map Spotify's engineered audio features to listening motives and cultural trends.

The papers below illustrate common methods (statistical analysis, classic ML, and multimodal / deep learning), datasets (Spotify API, Million Song Dataset, SpotGenTrack/SpotGen datasets, Kaggle collections, and YouTube-derived augmentations), and the remaining gaps (model explainability, temporal dynamics, and integration of visual/social signals).

1) *"A multimodal end-to-end deep learning architecture for music popularity prediction."*

Proposes **HitMusicNet**, a multimodal deep network that fuses audio, lyrics, and metadata via autoencoders to predict popularity. Uses the SpotGenTrack/SpotGen popularity dataset (audio + lyrics + Spotify popularity) and reports improved accuracy over single-modality baselines. Relevant because it shows multimodal inputs (not only Spotify audio features) improve prediction and highlights the value of deep learning for complex feature interactions.

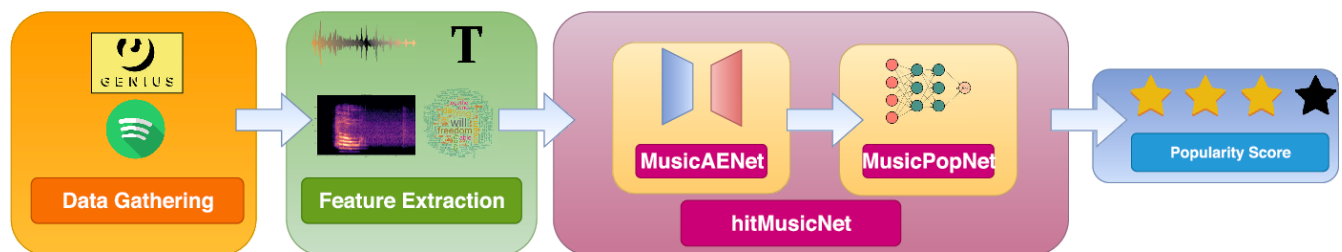


Fig 1: An illustrative High-level block diagram showing the pipeline of the presented study that attempts to promote the performance when predicting the popularity of a song.

2) “Effect of Feature Selection on the Accuracy of Music Popularity Classification Using Machine Learning Algorithms.”

Examines how feature-selection methods affect classification performance on Spotify-style datasets; compares filter- and wrapper-based feature selection with algorithms like Random Forest and SVM.

Dataset: Spotify audio features collected and preprocessed for classification of popularity bins. Relevant because it quantifies which subsets of Spotify features carry most predictive signal and stresses careful feature selection to avoid overfitting.

3) “Music we move to: Spotify audio features and reasons for listening.”

A descriptive study that links Spotify’s audio features (danceability, energy, valence, tempo, loudness, etc.) with listening motives and playlist contexts using large Spotify/playlist samples.

Method: statistical analysis and survey-linked interpretation (feature distributions, comparisons across playlist types). Relevant because it validates the interpretability of Spotify’s audio features and provides domain grounding for hypotheses about which features should correlate with popularity.

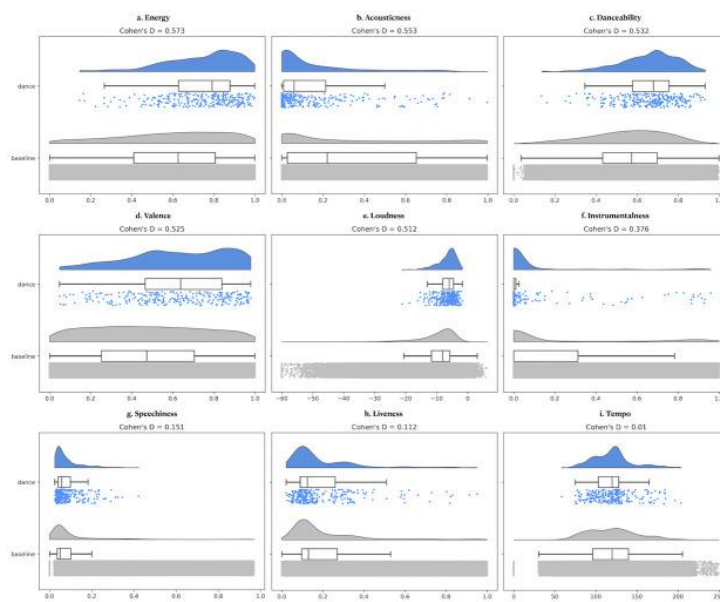


Fig 2: Audio features visualizations in PMC research paper

4) *“Evolution of global music trends: An exploratory analysis of Spotify features.”*

Performs an exploratory, cross-country study of popular tracks to identify macro trends (e.g., shifts in danceability/valence over time and across regions).

Dataset: Spotify charts across many countries and years, analyzed with descriptive statistics and trend visualizations. Relevant because it demonstrates temporal and cultural variability in feature-popular relationships — a notable limitation for static models trained on pooled data.

Prior work consistently shows Spotify’s engineered audio features are useful predictors and interpretable descriptors of listening behavior, while multimodal and social-signal integrations (lyrics, video, YouTube metrics) typically improve model performance.

Remaining gaps include:

- (a) **temporal dynamics** (how feature–popularity relations change over time)
- (b) **model explainability** (interpretable measures of feature importance across genres/cultures)
- (c) **dataset heterogeneity** (different popularity definitions and imbalanced labels).

This project can contribute by focusing on interpretable EDA, hypothesis tests per feature, and a baseline predictive model with clear, reproducible preprocessing bridging descriptive and predictive threads.

METHODOLOGY

This section outlines the systematic process followed for data acquisition, preprocessing, and analysis. The methodology ensures that the dataset used is clean, dependable, and suitable for statistical and predictive modeling.

(DATA COLLECTION & PREPROCESSING)

The dataset used in this project is the **Spotify Tracks Dataset**, obtained from **Kaggle**, which contains detailed information about a wide range of songs available on Spotify.

```
df = pd.read_csv("../Datasets/Spotify.csv")
```

The dataset includes approximately **[more than 1 lakh]** tracks, each described by multiple **audio and metadata features** such as,

"danceability, energy, loudness, tempo, key, valence, acousticness, speechiness, duration, popularity etc.."

To perform EDA, we need to ensure analytical accuracy. Several preprocessing steps were considered:

- **Duplicate Removal:** Identified and removed duplicate records to maintain data integrity.

```
df.drop_duplicates(inplace=True)
```

- **Missing Value Handling:** Checked for and treated missing or null entries in essential columns.

```
df.fillna("Unknown", inplace=True)
```

- **Outlier Treatment:** Detected and handled extreme values to prevent distortion of analysis results.

```
num_df = df.select_dtypes(include='number')

Q1 = num_df.quantile(0.25)
Q3 = num_df.quantile(0.75)
IQR = Q3 - Q1

outliers = ((num_df < (Q1 - 1.5 * IQR)) | (num_df > (Q3 + 1.5 * IQR))).sum()
print("Outliers per numeric column:\n", outliers)
```

- **Data Type Correction:** Ensured all variables were stored in appropriate formats (e.g., numeric for continuous data, categorical for labels).

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 113550 entries, 0 to 113549
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   track_id              113550 non-null  object
1   artists               113550 non-null  object
2   album_name            113550 non-null  object
3   track_name            113550 non-null  object
4   popularity             113550 non-null  int64
5   duration_ms           113550 non-null  int64
6   explicit              113550 non-null  bool
7   danceability           113550 non-null  float64
8   energy                113550 non-null  float64
9   key                   113550 non-null  int64
10  loudness              113550 non-null  float64
11  mode                  113550 non-null  int64
12  speechiness           113550 non-null  float64
13  acousticness          113550 non-null  float64
14  instrumentalness       113550 non-null  float64
15  liveness              113550 non-null  float64
16  valence               113550 non-null  float64
17  tempo                 113550 non-null  float64
18  time_signature         113550 non-null  int64
19  track_genre            113550 non-null  object
dtypes: bool(1), float64(9), int64(5), object(5)
memory usage: 16.6+ MB
```

- **Feature Refinement:** Converted duration from milliseconds to minutes for interpretability. Explicit boolean values were narrowed down to numeric 0 and 1 for easy interpretation and working.

```
df['duration_ms'] = (df['duration_ms'] / 60000).round(2)
df.rename(columns={'duration_ms': 'duration_min'}, inplace=True)
df['explicit'] = df['explicit'].astype(int)
print(df['explicit'].value_counts())
```

(TOOLS AND TECHNOLOGIES USED)

The project was implemented in **Python** using the **Jupyter Notebook extension in Visual Studio Code (VS Code)** as the development environment. This setup provided an interactive workspace for coding, visualization, and documentation, enabling efficient analysis and iterative testing.

The tools and libraries used are categorized below based on their functionality:

1. Core Data Handling and Computation

- **Pandas:** For loading, cleaning, transforming, and managing tabular data in DataFrames.
- **NumPy:** For numerical computations, array operations, and mathematical transformations.

2. Data Visualization

- **Matplotlib:** For creating static and customized visualizations such as bar charts, histograms, and scatter plots.
- **Seaborn:** For advanced and aesthetically enhanced statistical visualizations, including heatmaps, pairplots, and boxplots.
- **RocCurveDisplay** and **PrecisionRecallDisplay (from sklearn.metrics):** For visualizing model performance metrics such as ROC and precision-recall curves.

3. Data Preprocessing and Scaling

- **StandardScaler (Scikit-learn):** For feature normalization and standardization to ensure all numerical features contribute equally to analysis and modeling.
- **SMOTE (from imblearn):** For balancing the dataset by synthetically oversampling the minority class during classification.

4. Dimensionality Reduction and Clustering

- **PCA (Principal Component Analysis):** For reducing feature dimensionality and identifying key patterns in the data.
- **KMeans (Scikit-learn):** For unsupervised clustering of tracks based on similar audio characteristics.

5. Machine Learning and Predictive Modeling

- **RandomForestClassifier:** Used to build a predictive model that classifies songs as *hit* or *non-hit* based on their features.
- **train_test_split (Scikit-learn):** For dividing data into training and testing sets to evaluate model performance.
- **joblib:** For saving and reloading trained machine learning models efficiently.

6. Model Evaluation Metrics

- **accuracy_score, f1_score, roc_auc_score, classification_report, confusion_matrix, ConfusionMatrixDisplay.**
These functions were used to evaluate classification performance using multiple metrics for robustness and interpretability.

7. Statistical and Hypothesis Testing

- **SciPy (stats module):** For performing statistical hypothesis tests such as *t-tests* and *ANOVA* to validate feature significance.
- **Statsmodels (api & ols):** For conducting linear models and detailed statistical analysis.
- **MANOVA (Multivariate Analysis of Variance):** For testing the influence of multiple features on song popularity simultaneously.

IMPLEMENTATION

Kaggle's Spotify Tracks Dataset contains both categorical and numerical columns which specify the details about the tracks and their musical aspects in a refined way. Let us first understand the different columns and what exactly it means so that it becomes easy to provide future insights based on finding relations.

TRACK INFORMATION

Column	Description	Data Type
track_id	Unique identifier assigned to each song	Alphanumeric
artists	Performer(s) of the track	Text
album_name	The album in which the song appears.	Text
track_name	The title of the song.	Text
track_genre	The genre or category of the track	Text

CONTENT INFORMATION

Column	Description	DataValue
popularity	Popularity score (0–100) based on Spotify streams and listener engagement.	0 – 100
duration_ms	Total length of the track in milliseconds.	~100,000 – 500,000 ms
explicit	Indicates if the track contains explicit or adult content (True = explicit, False = clean).	Boolean

AUDIO FEATURES

Column	Description	Typical Values
danceability	How suitable a track is for dancing based on rhythm, tempo, and beat strength.	0.0 – 1.0
energy	Intensity and activity level of a song; higher values indicate more energetic tracks.	0.0 – 1.0
key	The musical key the song is in (0–11, representing C–B).	Integer (0–11)
loudness	Average loudness of the track in decibels (dB).	–60 to 0 dB
mode	Indicates if the track is in a major (1) or minor (0) key.	Binary (0 or 1)
speechiness	Detects spoken words or speech-like elements.	0.0 – 1.0
acousticness	Confidence measure of how acoustic a track is.	0.0 – 1.0
instrumentalness	Predicts the likelihood of a track having no vocals (closer to 1 = instrumental).	0.0 – 1.0
liveness	Probability that the track was recorded live (higher = live performance).	0.0 – 1.0
valence	Musical positivity or emotional tone (0.0 = sad, 1.0 = happy).	0.0 – 1.0
tempo	Estimated speed of the song in beats per minute (BPM).	50 – 220 BPM
time_signature	Estimated number of beats per bar in the song	Typically 3, 4, or 5

After getting an insight of what different column conveys, we moved into our first step of Exploratory Data Analysis that is **Data Cleaning**

Data Cleaning enhances the overall quality, accuracy, and reliability of the dataset, ensuring that any insights or models built upon it are meaningful and trustworthy. By removing duplicates, handling missing values, and correcting inconsistencies, it minimizes noise and bias in analysis. Outlier handling and proper scaling prevent extreme values from skewing results, while standardized formats make data easier to visualize and interpret. Clean data improves model performance, increases interpretability, and builds a solid foundation for advanced analysis such

as PCA, clustering, and predictive modeling — leading to more confident and actionable decisions.

For this analysis, the different Data Cleaning steps taken were:

- Dropping duplicates
- Filling missing values
- Checking consistency of values for the permissible values
- Converting milliseconds to minutes for easy understanding
- Normalizing categorical columns like “explicit”
- Outlier detection

There were a lot of outliers but for this we did not remove them. A song has its own unique characteristics and so its features might not follow the same trend as others and thus might not lie between the lower and upper bound of the column. This is perfectly fine and when building model, it gives it a wide variety of characteristics to look upon.

Now our data is ready, and we are moving onto the next step that is **Data Exploration**

Data Exploration and Preprocessing are crucial steps that bridge raw data with meaningful insights and dependable models. They allow us to understand the structure, quality, and relationships within the dataset before any modeling begins.

Exploration helps uncover hidden trends, correlations, and data imbalances, while preprocessing ensures the dataset is consistent, normalized, and ready for machine learning. Together, they enhance accuracy, interpretability, and performance, reducing the risk of misleading results and improving generalization on unseen data.

Lets us uncover the insights of our analysis and dataset

```
print(f"No of rows: {df.shape[0]}")  
print(f"No of columns: {df.shape[1]}")
```

```
No of rows: 114000  
No of columns: 20
```

The dataset contains 114000 x 20 cells. These 20 cells consist of both numerical and categorical data. All these data together contribute to the musical aspect of a song.

	# popularity	# duration_ms	# danceability	# energy	# key	# loudness
count	113550.0	113550.0	113550.0	113550.0	113550.0	113550.0
mean	33.32413914575077	228079.36220167327	0.5670307151034787	0.6420903408824307	5.309467195068252	-8.243418705
std	22.28397609446146	106414.78255370946	0.17340845184391565	0.25105223813389455	3.5601344693837897	5.011400798
min	0.0	0.0	0.0	0.0	0.0	0.0
25%	17.0	174180.25	0.456	0.473	2.0	-9.0
50%	35.0	213000.0	0.58	0.685	5.0	-8.0
75%	50.0	261587.75	0.695	0.854	8.0	-7.0
max	100.0	5237295.0	0.985	1.0	11.0	-1.0

Using Pandas, `df.describe` we get a descriptive analysis of the different columns which help us understand the various aspects.

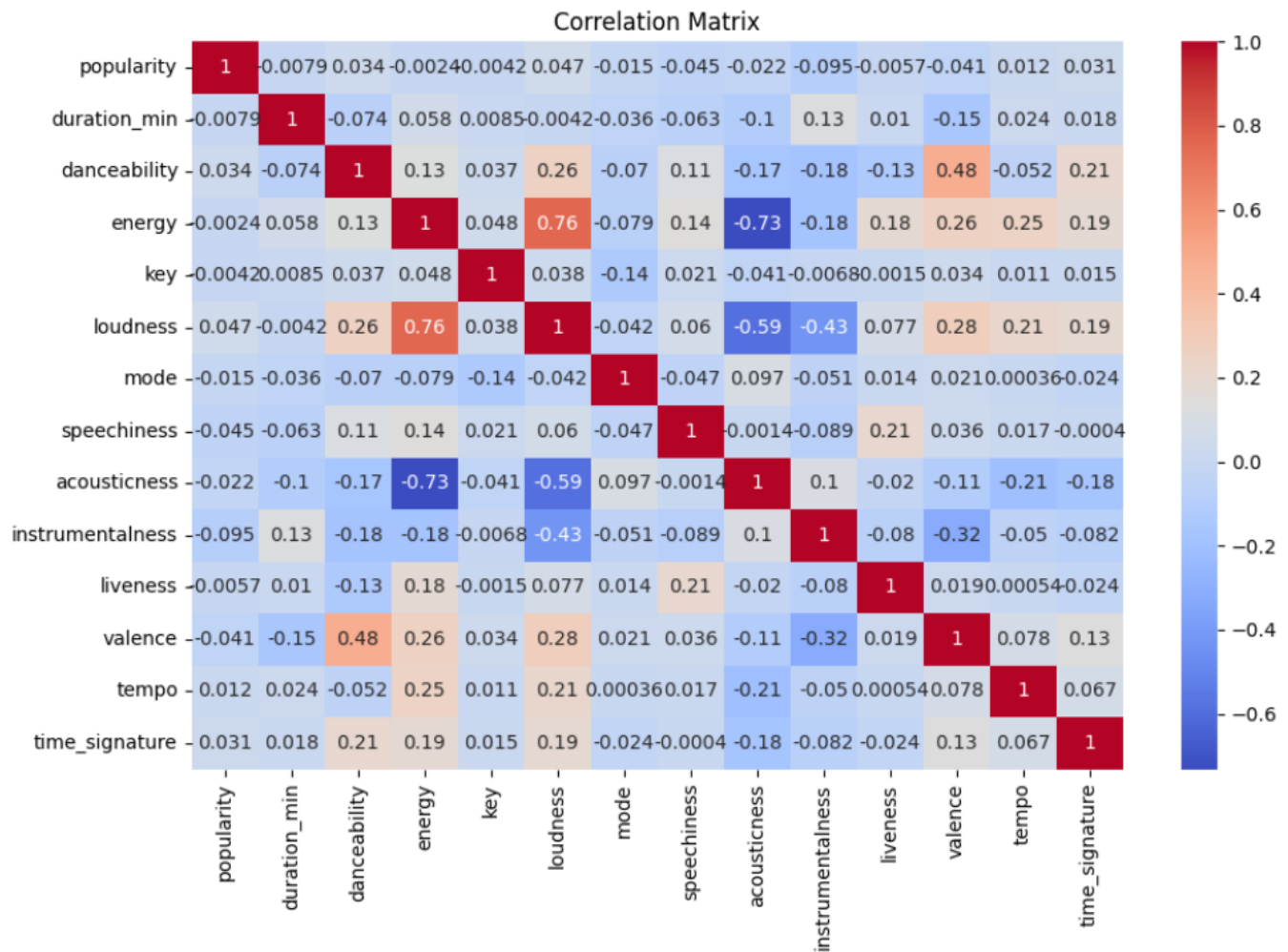


Fig 3: Correlation Matrix to show relationships

Plotting a Correlation Matrix helps discover relations between different numerical columns and see which columns are related the most either Positively or Negatively. The red ones signify the positive and the blue ones signify the negative correlations

In the given matrix we can see:

- **Loudness** and **Energy** (Correlation **0.76**) is **Strongly Positively** correlated signifying that loud music contributes to more energy in people.
- **Acousticness** and **Energy** (Correlation **-0.73**) is **Strongly Negatively** correlated signifying that songs that are more acoustic (softer) tend to have less energy

Univariate Analysis helps understand each column separately which helps understand its spread, descriptive analysis, variation in data.

One such univariate analysis is understanding how skewed the data is. Understanding about it helps understand how the data is and what exactly its representation conveys.

Feature	Skew Value	Skewness Description	Meaning / Insight
duration_ms	10.81	Highly right-skewed	Most songs are short, but a few exceptionally long tracks stretch the tail.
speechiness	4.64	Strong right-skewed	Few spoken word/rap tracks have high values; most songs have low speech content.
explicit	2.96	Right-skewed (binary)	Majority of songs are non-explicit (0). Skewness is not meaningful for binary.
liveness	2.10	Right skewed	Most songs are studio-recorded; only a few are live performances.
instrumentalness	1.73	Right skewed	Most songs are vocal; only a few are fully instrumental.
acousticness	0.73	Slightly right skewed	Acoustic-heavy songs are fewer.
tempo	0.23	Nearly symmetrical	Good variety of song tempos.
valence	0.11	Nearly symmetrical	Emotional positivity (valence) is balanced.
popularity	0.04	Symmetrical	Popularity is evenly distributed.

key	-0.008	Symmetrical	Categorical variable (0–11). Ignore skewness.
danceability	-0.40	Slightly left-skewed	Most songs are moderately danceable.
mode	-0.57	Left-skewed (binary)	Major/minor mode — skewness not meaningful for binary.
energy	-0.59	Slightly left-skewed	Presence of some low-energy tracks.
loudness	-2.01	Left-skewed	Most tracks are loud; few quiet tracks pull the tail left.
time_signature	-4.11	Strong left-skewed	Most songs use 4/4; unusual time signatures are rare.

Outliers also form a part of the dataset and represent unusual data which are significantly different from other data in the columns. We have used the IQR (Interquartile-range method) to determine whether the data points are capped between the lower and upper bounds.

Column	Outlier Count	Outlier Level	Interpretation
popularity	2	Very few	Popularity scores are mostly consistent — Spotify’s 0–100 range naturally limits extremes.
duration_ms	5,593	Moderate	Some tracks are much longer or shorter than typical songs (e.g., intros, podcasts, extended versions).
explicit	9,718	Moderate	Many songs are non-explicit (0), fewer are explicit (1). Binary — so these aren’t true numeric outliers.
danceability	615	Few	Distribution is consistent — small deviations are acceptable.
energy	0	None	No extreme deviations — energy values remain consistent.
key	0	None	Musical key (0–11) is discrete — outlier detection doesn’t apply.

loudness	6,115	Moderate	Some tracks are unusually quiet or loud — could influence energy perception.
mode	0	None	Binary (0/1) — outlier concept not meaningful.
speechiness	13,167	High	Spoken-word, rap, or podcast-style tracks cause extremes — expected in mixed datasets.
acousticness	0	None	No significant anomalies — distribution is stable.
instrumentalness	25,131	Very High	Many tracks are vocal (0) while a few are 1.0 (pure instrumentals). Expected in music datasets.
liveness	8,610	High	Live performances create natural variation — high outlier count is expected.
valence	0	None	Well-balanced values — no extreme emotional-tone deviations.
tempo	615	Few	A few tracks are extremely fast/slow — could be remixes, classical, or niche genres.
time_signature	12,063	High	Most songs are 4/4; uncommon signatures (3/4, 5/4) appear as outliers — musically interesting.

Removing these outliers does not make sense for this dataset, as these values are not weird but signify the characteristic for their songs which are unique for every song and depends upon the artists.

After analyzing and performing univariate analysis and understanding the outliers, we can now move on to one of the most important aspects of our dataset, **FEATURE ENGINEERING**. This will allow us to uncover patterns and find/create new data columns from the existing ones allowing us to understand how the musical aspects of data contribute to the mood, vibe, etc.

Feature engineering involves combining multiple columns and applying certain formulas to their values to find a new feature. This also helps remove noise and unnecessary features to reduce model complexity

We applied feature engineering and created 10 new columns from the existing data using certain formulas listed in the table below.

Feature Name	Formula	Interpretation
popularity_level	Categorized using bins: Low (0–40), Medium (40–70), High (70–100)	Converts numeric popularity into meaningful categories for analysis.
duration_category	Short (0–2.5 min), Medium (2.5–4.5), Long (4.5–15)	Groups songs by length to compare behavior of different duration types.
mood_intensity	$(\text{valence} \times \text{energy}) - (\text{loudness} / 60)$	Measures emotional strength of a song; higher = more intense mood.
replay_value_score	$0.35 \times \text{dance} + 0.35 \times \text{energy} + 0.20 \times (1 - \text{instrumental}) + 0.10 \times (\text{ideal duration})$	Predicts how likely a song is replayed; higher = more replay-worthy.
party_index	$0.4 \times \text{energy} + 0.4 \times \text{dance} + 0.2 \times (1 - \text{speechiness})$	Higher = better for parties (energetic, danceable, low vocals).
sadness_score	$(1 - \text{valence}) \times (1 - \text{energy})$	Higher = sad, low-energy tracks; useful for mood-based playlisting.
production_complexity	$0.5 \times \text{instr} + 0.3 \times \text{speech} + 0.2 \times \text{liveness}$	Higher = more complex production elements in the track.
artist_popularity_deviation	$\text{popularity} - \text{mean}(\text{popularity per artist})$	Shows if a song performs above/below the artist's usual level.
vibiness_index	$0.4 \times \text{dance} + 0.3 \times \text{valence} + 0.3 \times (\text{tempo in } 90\text{--}140 \text{ bpm})$	Measures “vibe” factor; mid–high tempo, feel-good, danceable tracks score high.
skip_likelihood	Based on long duration, low energy, low dance, high instrumentalness	Estimates how likely a user may skip the track. Higher = more likely to skip.

These new columns help us understand dataset more effectively, which forms the basis of our next step that is **Data Reduction and Model Building**. It has also performed the multivariate analysis uncovering the patterns hidden in the dataset.

Principle Component Analysis (PCA) is a data reduction technique used to **compress many features into fewer “principal components”** while keeping maximum information. This will help summarize many correlated features, remove noise from skewed features and most importantly, prepare features for machine learning.

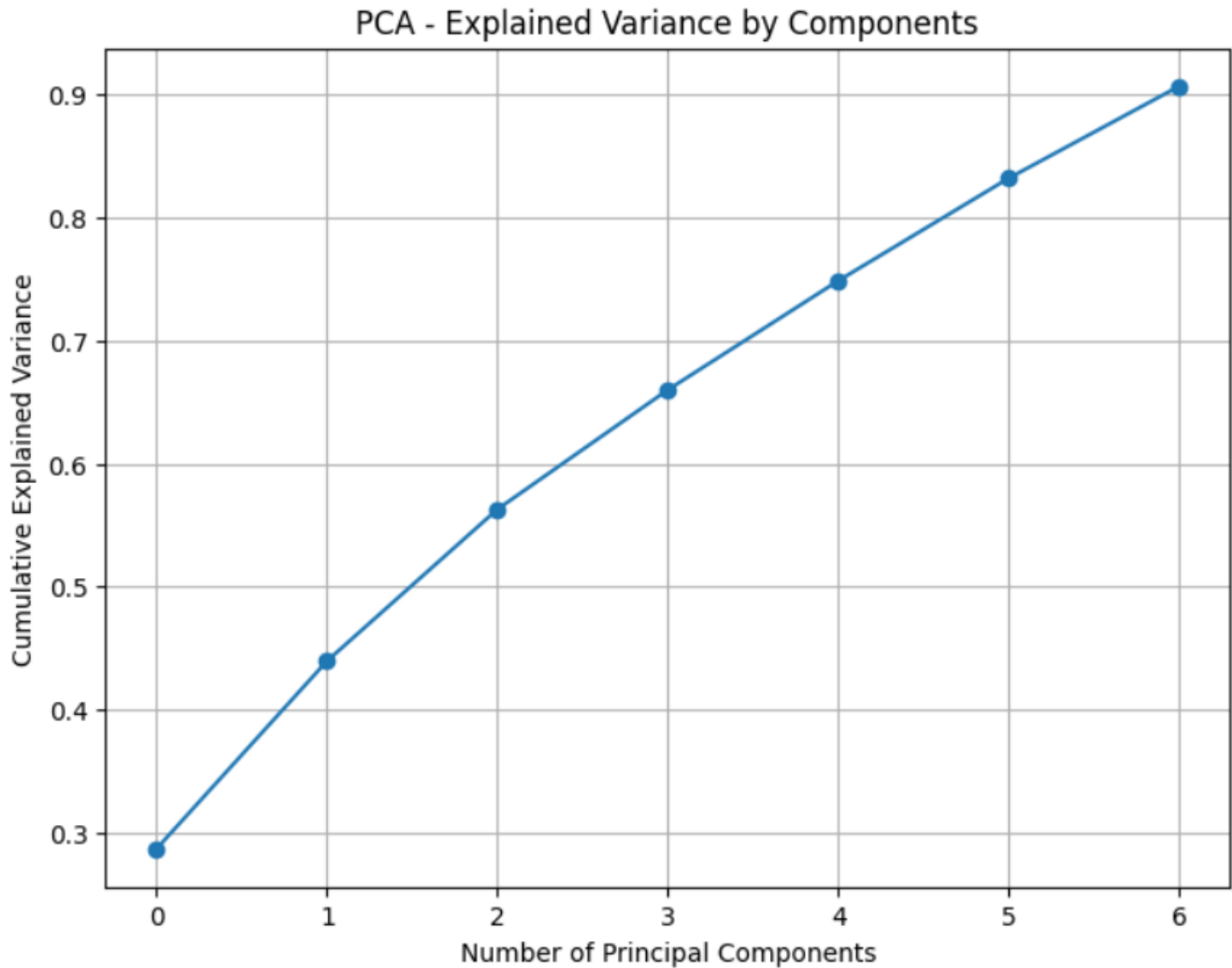
We need to choose only the numeric data for this and standardize it and then apply PCA using Sckit-Learn’s inbuilt PCA module. This created 7 new columns namely PC1, PC2, PC3..etc. We also plotted the result for better understanding.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
duration_min	-0.015	-0.404	-0.173	-0.576	-0.039	0.667	-0.142
loudness	0.517	-0.077	-0.039	0.039	-0.183	-0.077	-0.167
danceability	0.245	0.484	-0.219	-0.391	0.270	0.034	0.137
energy	0.509	-0.263	0.038	-0.056	-0.004	-0.190	0.091
speechiness	0.101	0.092	0.609	-0.259	0.562	-0.089	-0.419
acousticness	-0.438	0.316	0.177	0.116	-0.010	0.294	-0.036
instrumentalness	-0.283	-0.322	-0.153	-0.237	0.456	-0.334	0.532
liveness	0.085	-0.142	0.696	-0.059	-0.265	0.184	0.537
valence	0.296	0.487	-0.074	-0.013	0.060	0.280	0.414
tempo	0.189	-0.240	-0.067	0.609	0.543	0.440	0.069

Fig 4: PCA component loadings showing contribution of original features to each new principal component.

Principal Component	Best Interpreted As	Description
PC1	Energy & Production Intensity	High-energy, loud, studio-produced tracks ↔ soft, acoustic, calm songs
PC2	Danceability & Positivity (Feel-Good Vibe)	Party/dance, upbeat, happy tracks ↔ ambient, emotional, low-vibe tracks
PC3	Vocal Presence & “Live” Performance Factor	Spoken words, rap, strong vocals, or live recordings
PC4	Tempo vs. Song Length Balance	Fast, short, energetic songs ↔ slow, long ballads
PC5	Speech + Pace Intensity	Faster vocal-heavy tracks (rap / spoken-fast)
PC6	Song Duration Dimension	Distinguishes long vs short tracks (duration-driven)
PC7	Live Instrumental Blend	Acoustic, concert, or unplugged-style tracks vs studio-processed

We also plotted a graph showing how much total information is preserved as we increase the number of principle components



***Fig 5:** Graph showing how cumulative explained variance increases with the number of principal components.*

This graph shows the **cumulative explained variance** from **Principal Component Analysis (PCA)**.

- The **x-axis** represents the number of principal components included.
- The **y-axis** shows how much of the total data variance is explained by those components combined.
- The curve rises as more components are added, meaning each new component explains additional variance.
- Around the 5th or 6th component, the curve starts to **flatten**, indicating **diminishing returns** — most of the data's information is already captured.

MODEL BUILDING is an integral part of our Exploratory Data Analysis. After getting what our dataset contains and conveys, we will formulate a model which correlates to our Problem Statement to distinguish a song as “**HIT**” or “**NOT HIT**” for future data.

Since we performed feature engineering and data reduction, we will now use them to train and run a model using Random Forest Classifier with key hyperparameters

- *Estimators*: 500 trees
- *Max Depth*: 18
- *Class Weight*: balanced_subexample
- *Min Samples Split*: 5
- *Min Sample Leaf*: 2

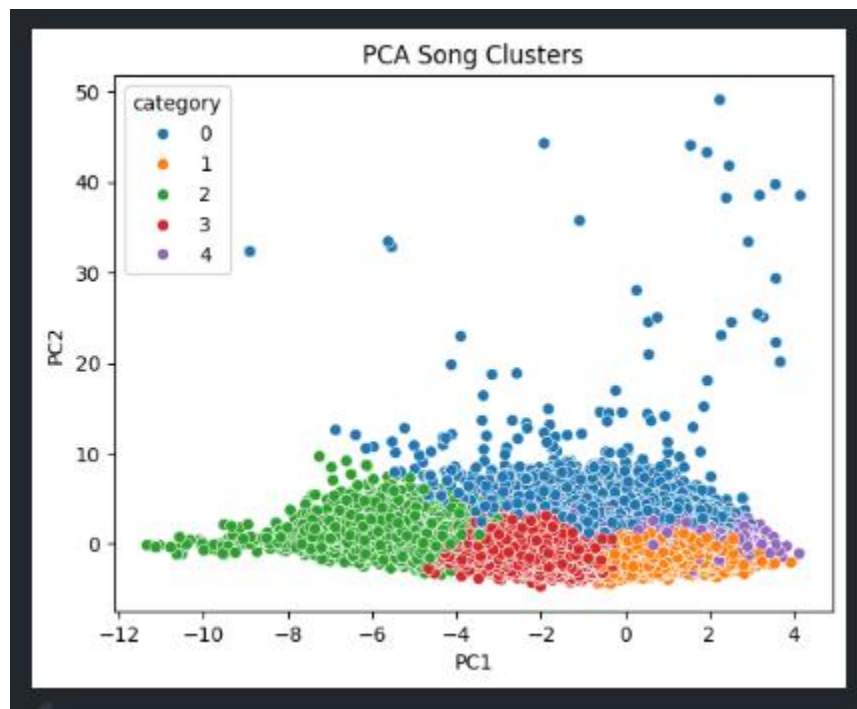
Hit vs Non-Hit distribution was imbalanced so we applied, **SMOTE (Synthetic Minority Oversampling Technique)** to identify better hits instead of being biased towards non-hit.

Using a probability decision threshold of **0.35** (instead of default 0.5), the model improves hit-detection sensitivity.

The results included:

- Accuracy: 0.599
This tells 59.9% predictions were correct
- F1 Score: 0.66
Strong Balanced between precision & recall for hit detection
- ROC AUC: 0.805
High ability to distinguish hit vs non-hit songs

The current model focuses on **maximizing the detection of hit songs**, achieving a strong ROC-AUC of **0.805** and excellent recall of **0.95** for hits. While accuracy is moderate due to false positives, this trade-off supports the objective of capturing likely hits for music recommendation, marketing, and A&R decision-making.



***Fig 6:** Scatter plot showing clusters of songs based on PC1 and PC2, the most significant principal components.*

The above visualization shows the results of applying **K-Means clustering (k=5)** on the PCA-reduced song features. Each point represents a song plotted on the first two principal components, and each color indicates a cluster of songs that share similar musical characteristics.

The spread and separation of clusters indicate that PCA successfully captured meaningful patterns in the audio features, allowing the model to group songs into distinct categories such as high-energy tracks, acoustic/low-energy songs, balanced mid-tempo tracks, and other niche styles.

This clustering helps us understand natural groupings in the dataset and can be used for **song categorization, playlist generation, and personalized recommendation systems.**

After running several sample tests, and analyzing the predictions, we formulated a table which defines a permissible value for a song to go **Hit** or not.

Feature	Hit Song Characteristics	Non-Hit Characteristics	Insight / Why It Matters
Danceability	0.70–0.90 (High)	<0.40 (Low)	Catchy rhythmic tracks are easier to replay and share.
Energy	0.80–1.00 (High)	<0.50 (Low)	Energetic, lively music aligns with mainstream pop/EDM trends.
Valence	0.60–0.90 (Positive)	<0.40 (Sad/Low)	Happier sounding tracks gain wider audience acceptance.
Loudness	–6 dB to –3 dB (Loud)	Quieter than –12 dB	Loud, well-mastered songs stand out sonically and feel more impactful.
Acousticness	Low (<0.30)	High (>0.60)	Hits are typically digitally produced; acoustic tracks stay niche.
Instrumentalness	Near Zero	>0.40 (High)	Vocals drive emotional and lyrical engagement—key for virality.
Tempo (BPM)	110–130 BPM	<80 or >150 BPM	Mid-tempo songs align with dance/pop pacing and feel more universal.
Duration	2.5–4.0 min	<2 or >5 min	Optimal for streaming; short enough to replay, long enough to engage.
Speechiness	0.04–0.10 (Moderate)	>0.20 (High)	Slight rap/spoken elements work, but too many feel non-musical.
Liveness	0.10–0.30	>0.60 (Very High)	Studio-produced songs outperform live-recorded tracks commercially.
Mode (Musical Key)	Major (Mode = 1)	Minor (Mode = 0)	Major keys sound uplifting and mainstream; minor keys feel emotional/somber.

This can allow us to manually analyze whether a song based on its musical aspects will be popular or not. It might vary as well because these are emotional aspects and depend upon fame, trend on social media etc.

RESULTS

Now our Data Exploration and Model Building is perfectly completed, we can move onto our next step, **Data Visualization** to understand trends, data more easily using meaningful visuals.

Data Visualization helped us understand the Spotify data and distribution of various music features. It also became a key contributor when performing data reduction and model building to understand how the trend goes.

For our dataset, we segregated the columns into numeric and categorical to understand and choose the best plots for their visualization.

For numeric data we chose

- Histograms: To understand the distribution of data
- Box Plots: To detect outliers
- Bar Chart: To count frequency of distinct categories
- Pie Chart: To find percentage of frequency of distinct categories
- Scatter Plot: To show relation-based distribution across distinct categories

A few examples of visuals are given below

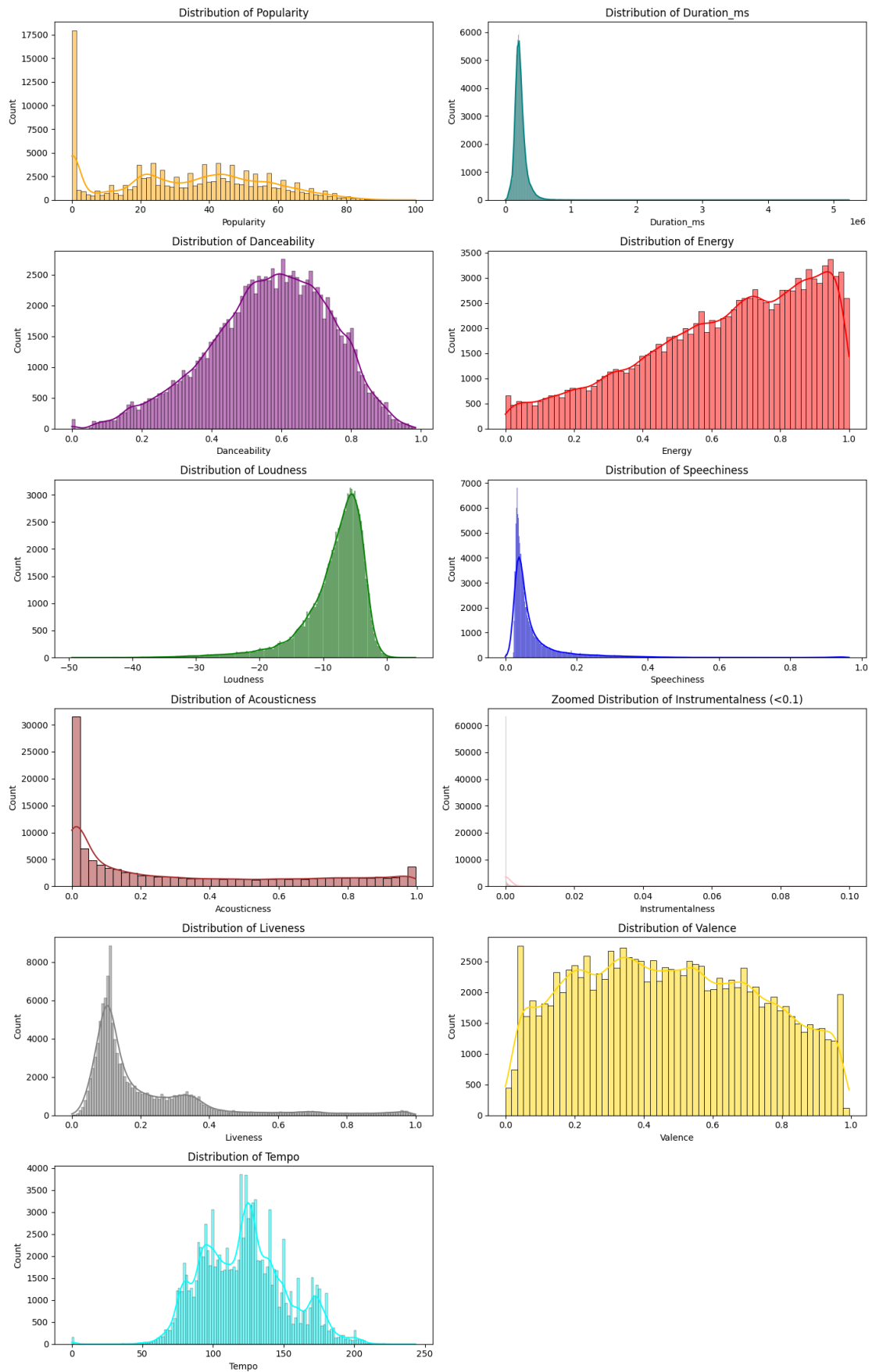


Fig 7: Histogram of univariate columns

Observations:

- Most tracks are energetic, studio-produced, and contain vocals rather than being acoustic or instrumental.
- Popular songs tend to be short, loud, and moderately danceable—fitting typical pop and radio standards.
- Emotional tone (valence) is balanced, showing both upbeat and mellow tracks coexist in popularity.
- Common tempos (90–150 BPM) align with mainstream genres like pop, dance, and electronic.

Spotify's music landscape is shaped by high-energy, electronically produced tracks optimized for streaming audiences. Songs are crafted for engagement—catchy, rhythmic, and studio-polished—while still spanning a wide emotional range, reflecting the diversity of listener preferences across global pop culture.

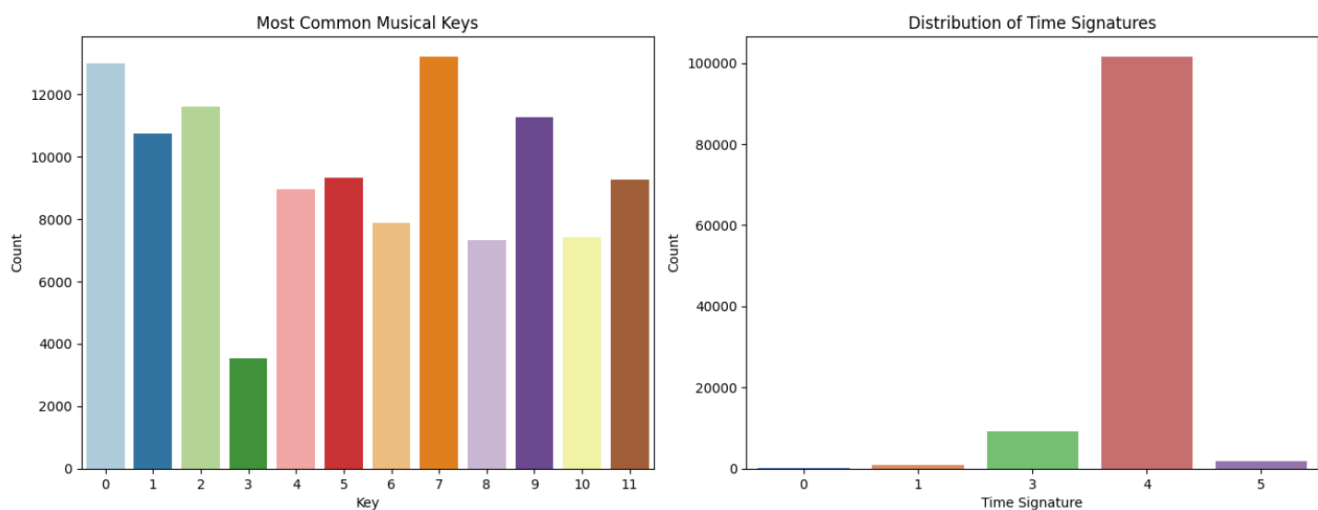


Fig 8: Dominance of 4/4 rhythm and common musical keys

Observations:

The analysis shows that most songs are composed in a few dominant musical keys, particularly keys 0, 2, and 7, indicating a preference for familiar tonal structures that appeal to a wide audience. The overwhelming majority of tracks use a 4/4-time signature, which is the most common rhythm in popular music due to its simplicity and danceable feel. This pattern suggests that mainstream tracks tend to follow conventional and listener-friendly compositions, emphasizing predictable rhythms and harmonic consistency that enhance accessibility and commercial success.

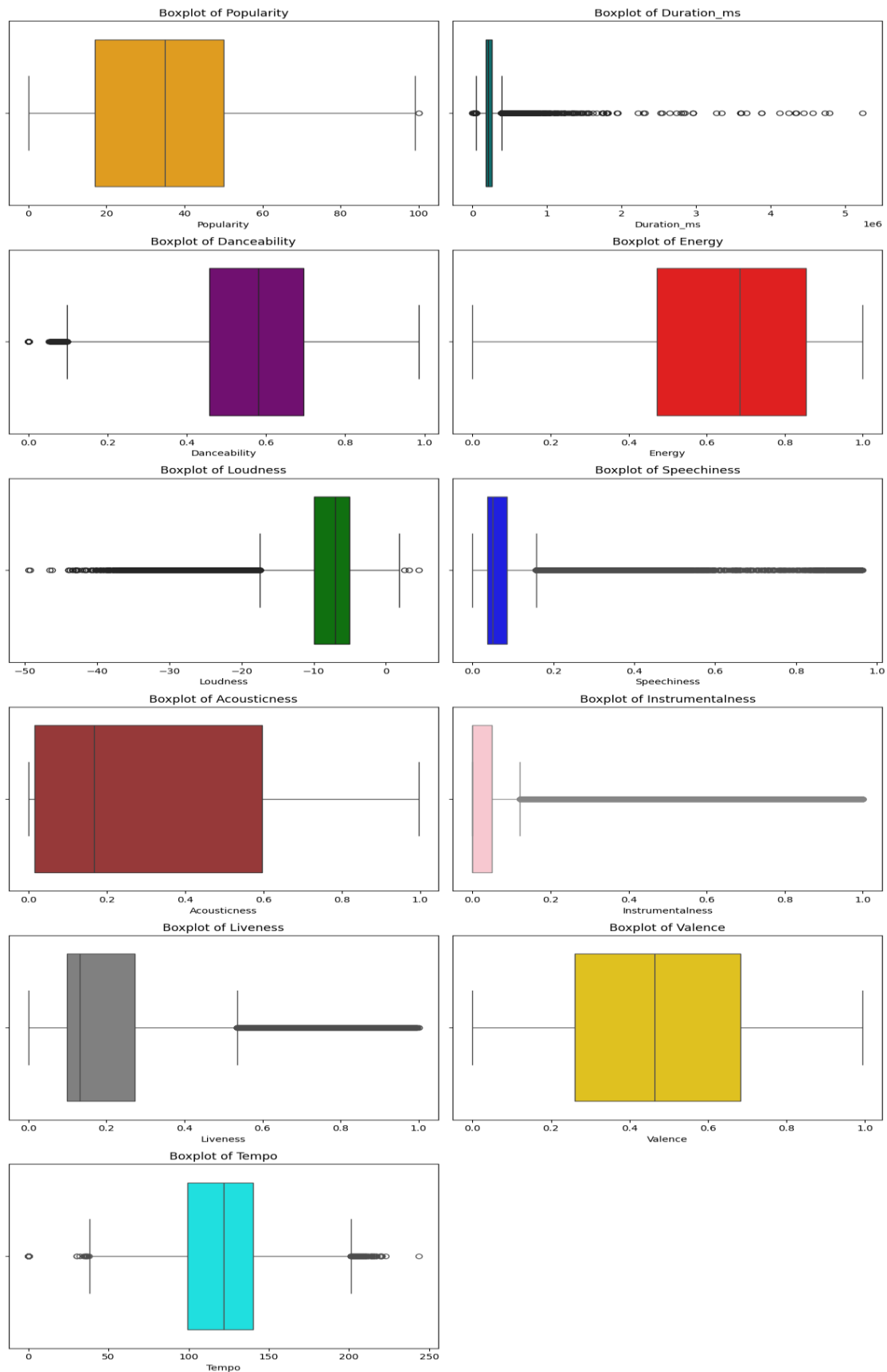


Fig 9: Boxplots showing spread, skewness, and outliers across audio features

Observations:

- Several features like *duration_ms*, *speechiness*, *instrumentalness*, and *liveness* contain numerous outliers, showing high variability across tracks.
- *Popularity*, *danceability*, *energy*, and *valence* have relatively symmetric distributions with moderate spread, suggesting these attributes are consistent among most songs.
- *Loudness* and *tempo* show few extreme values, likely representing remixes or unusual production styles.
- *Acousticness* and *instrumentalness* display strong right-skewness, confirming most tracks are non-acoustic and vocal-heavy.

The dataset exhibits significant diversity, with some audio features being consistent while others vary widely due to outliers and genre differences. This mix indicates a broad musical range on Spotify — from typical pop tracks to rare experimental or live recordings — making it suitable for analyzing how feature extremes might influence popularity.

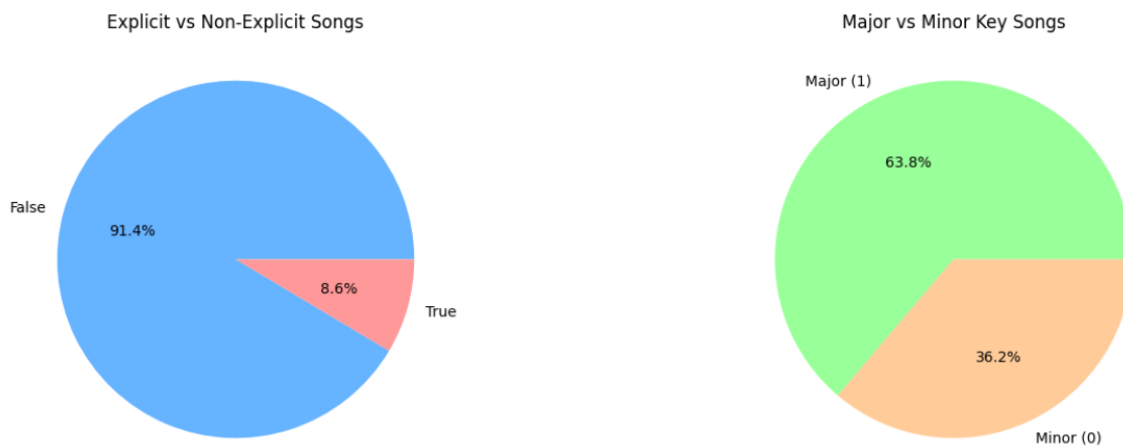


Fig 10: Distribution of explicit content and musical modes

Observations:

A deeper look reveals that Spotify's catalog is heavily oriented toward broad audience appeal. The dominance of non-explicit tracks (91.4%) suggests platforms and artists prioritize accessibility and radio-friendliness, aligning with mainstreaming standards. The higher proportion of major key songs (63.8%) further indicates a psychological trend toward positive, energetic, and emotionally uplifting compositions—qualities often associated with commercial success and listener retention. Together, these trends reflect how the modern music industry strategically favors emotionally upbeat, clean content that maximizes global reach and replay value.

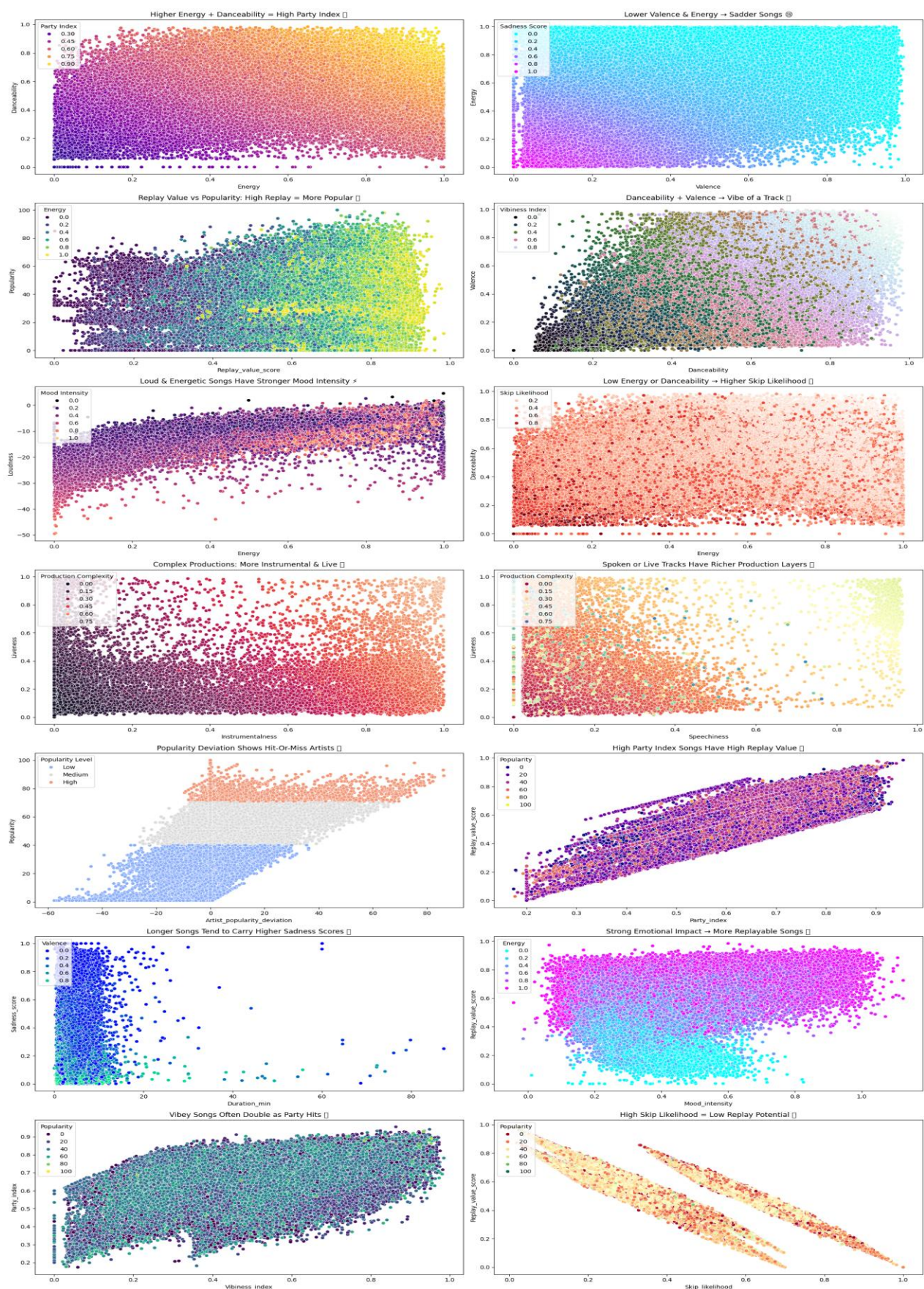


Fig 11: Scatter plots showing relationships between key audio features

Observations:

- **Energy & Loudness** (top left): Shows a strong positive correlation - higher energy tracks tend to be louder, with a clear linear gradient from purple (lower values) to yellow/orange (higher values).
- **Acousticness vs Danceability** (top right): These appear inversely related - highly acoustic tracks tend to have lower danceability scores, visible in the cyan cluster at high acousticness values.
- **Popularity Metrics** (middle left): Popularity shows significant spread across energy levels, suggesting that both high and low energy songs can be popular, though there's notable clustering.
- **Valence Relationships**: Multiple plots show valence (musical positivity) has complex, non-linear relationships with other features - it doesn't strongly predict other characteristics.

The Spotify dataset reveals that some audio features are strongly correlated (energy-loudness), while others like tempo and valence operate more independently, suggesting music's multidimensional nature in how it's categorized and perceived.

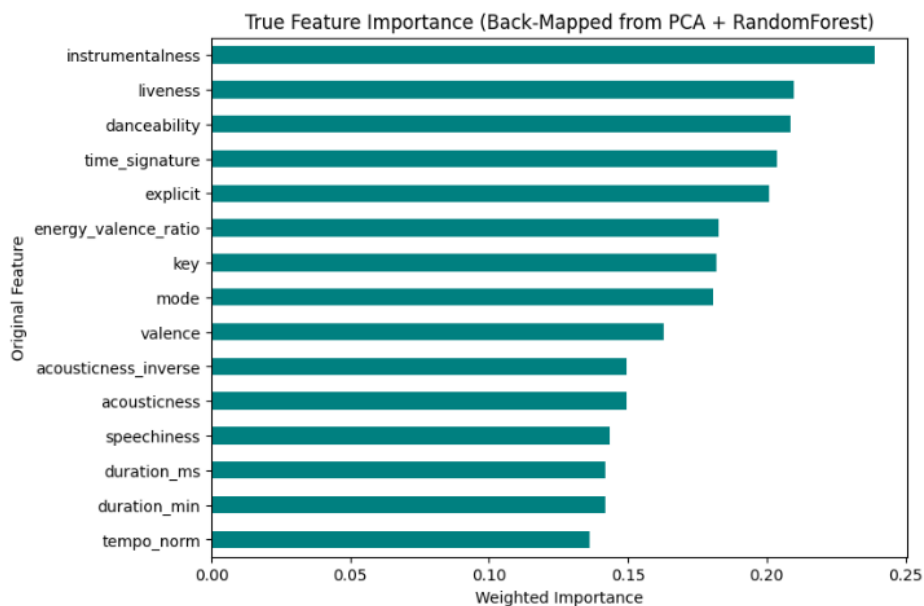


Fig 12: Feature importance scores showing the most influential song attributes used for modeling.

Feature importance ranking showing that vocal presence (low instrumentalness), rhythmic appeal (high danceability), and studio quality (low liveness) are the strongest predictors of a hit song, while tempo and duration have comparatively minor influence.

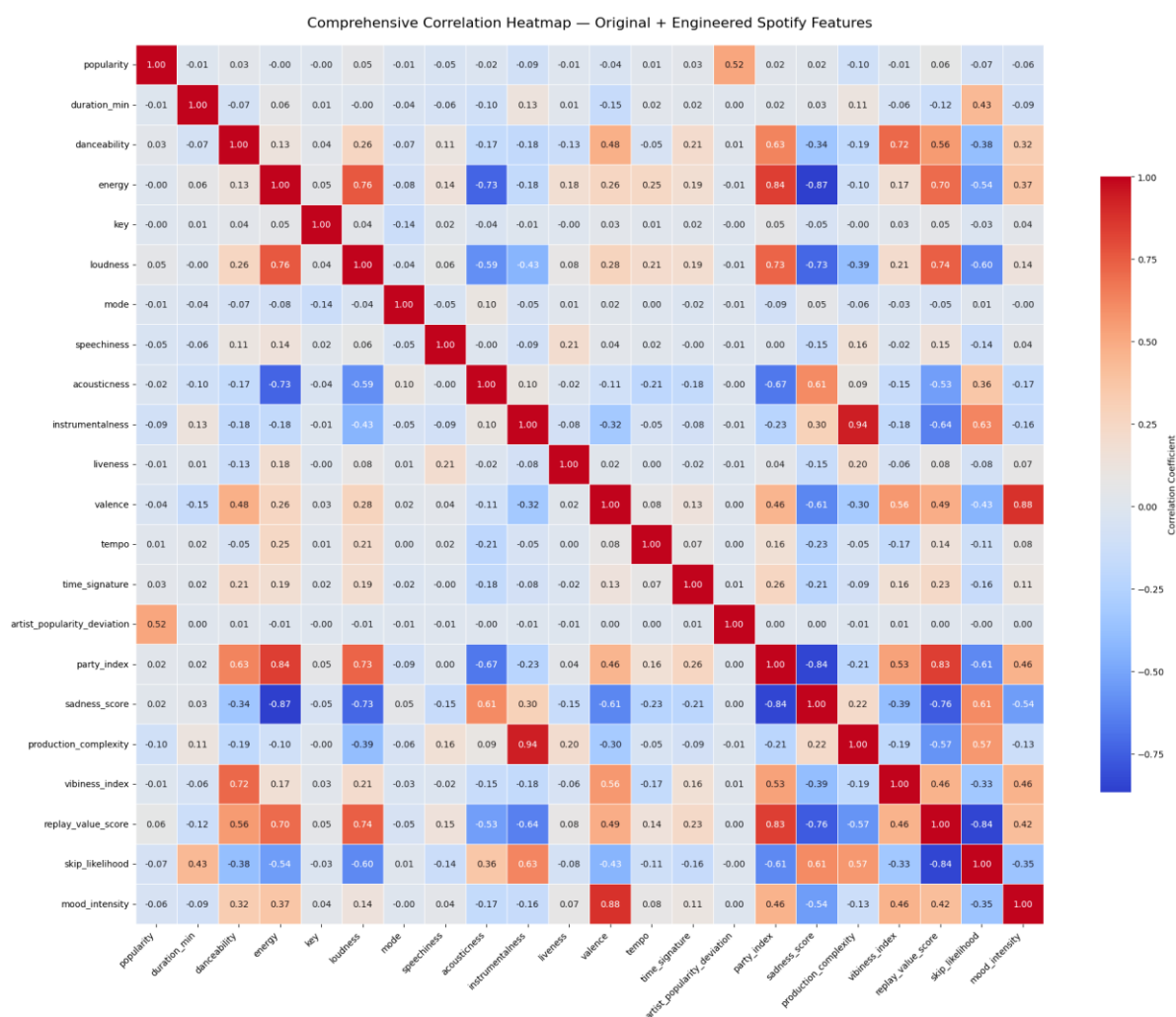


Fig 13: Comprehensive Correlation Heatmap of Spotify Audio Features

Strong Relationships:

- Audio features form clear clusters: energy-related features move together, mood/emotional features correlate, and acoustic properties oppose energetic ones
- Engineered metrics (party index, replay value, mood intensity) successfully capture combinations of underlying audio features

Weak Relationships:

- Popularity is largely unpredictable from audio features alone - hits can have any combination of characteristics
- Technical features like key, tempo, and duration operate independently

The dataset shows that music has two distinct correlation zones: **predictable acoustic relationships** (loud = energetic, acoustic = calm) and **unpredictable success factors** (popularity is independent of sound characteristics).

Having visualized the dataset, its now time to perform some **Hypothesis Testing** to verify our results and analyze them.

Hypothesis Testing will help determine whether the observed results in a dataset are **statistically significant** or occurred by **random chance**.

We formulated several questions along with their **Null** and **Alternate Hypothesis** and took help of Python libraries to use certain pre-defined testing methods to come to a conclusion.

Q1. Do explicit songs have a significantly different popularity than non-explicit songs?

- *H_0 : There is **no difference** in average popularity between explicit and non-explicit songs.*
- *H_1 : There **is a significant difference** in average popularity between explicit and non-explicit songs.*

```
t_stat, p_val = stats.ttest_ind(explicit, non_explicit, equal_var=False)
```

For this, we applied **t-test** to see whether explicit content affects song's success. The result came out to **reject null hypotheses** with **t-statistic (13.682)** which states that there is a statistically significant difference in popularity between explicit and non-explicit songs. This implies that the presence of explicit content has a measurable impact on how popular a song becomes on Spotify.

Q2. Does the average popularity differ significantly across track genres?

- *H_0 : The **mean popularity** is the same across all genres.*
- *H_1 : At least **one genre** has a **significantly different** mean popularity.*

```
anova_result = stats.f_oneway(*[group['popularity'].values for _, group in df.groupby('track_genre') if len(group) > 2])
```

For this, we applied **ANOVA** to see whether the mean is significantly different or not. The result came out to **Reject null hypothesis** with **F-Statistic (338.191)**. There is a statistically significant difference in average popularity across music genres. This suggests that **genre type meaningfully influences a song's popularity**.

Q3. Is there a significant relationship between a song's **energy** and its **popularity** on Spotify?

- H_0 : There is **no significant correlation** between a song's energy and its popularity.
- H_1 : There is a **significant correlation** between a song's energy and its popularity.

```
contingency = pd.crosstab(df['popularity_level'], df['explicit'])  
chi2, p_chi, dof, expected = stats.chi2_contingency(contingency)
```

For this we applied **Chi-Square Test** as we are comparing categorical column. The result came out to **Reject null Hypothesis** with **Chi² value (481.954)**. There is a statistically significant positive correlation between a song's energy and its popularity, indicating that high-energy tracks tend to be more popular on Spotify. This suggests that listeners are generally more drawn to **lively, dynamic songs** with **stronger intensity** and **upbeat characteristics**.

CONCLUSION and FUTURE WORK

This study explored the audio characteristics that influence whether a song becomes a hit on Spotify. Based on statistical analysis, feature engineering, PCA, and machine learning modeling, the findings show clear measurable patterns that differentiate hit songs from non-hits. Key indicators such as **energy, danceability, loudness, valence**, and optimal **song duration** emerged as strong contributors to a song's success.

The Random Forest model, enhanced with PCA and SMOTE, demonstrated the ability to predict hit songs with a **ROC-AUC of 0.805**, showing good discrimination capability. Clustering further revealed natural groupings of songs, providing insights into different music types and listener preferences. Overall, the project validates that hit-song prediction is feasible using audio features alone.

The scope of this study can be expanded to enhance the accuracy and depth of predictions. Potential future directions include:

- **Advanced Predicting model:**
Experiment with deep learning architectures such as LSTMs, Transformers, or CNNs to capture temporal and acoustic patterns.
- **Lyrics and Sentiment Analysis:**
Incorporate lyrical content, sentiment polarity, and emotion scoring to measure the impact of lyrics on song popularity.
- **Contextual Streaming Data:**
Integrate listener demographics, geographic trends, playlist inclusion, and temporal streaming behavior for richer context.
- **Marketing & Virality Factors:**
Consider social media trends, TikTok influence, release timing, and promotional campaigns.
- **Real-Time Prediction System:**
Deploy the model as an interactive web app or API to predict song success instantly for artists and producers.

This project lays a strong foundation for predictive music analytics. Expanding into lyrical, cultural, and behavioral data can significantly improve model performance and offer a more holistic understanding of what makes a song a global hit

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- Khan, A., Duru, O., & Kim, S. (2022). *Effect of feature selection on the accuracy of music popularity classification using machine learning algorithms*. Mathematics, 10(19).
- Duman, D., Fard, A. M., & Karlaftis, M. (2022). *Music we move to: Spotify audio features and reasons for listening*. PLOS ONE, 17(8).
- Terroso-Sáenz, F., Martínez-Ballesté, A., & García Gracia, M. (2023). *Evolution of global music trends: An exploratory analysis of Spotify features*. Journal of Retailing and Consumer Services, 72.

Dataset

Kaggle. (2023). *Spotify Tracks Dataset*.
<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>

Web Resources

- Spotify for Developers. (2023). *Spotify Web API Documentation*.
<https://developer.spotify.com/documentation/web-api/>
- Scikit-Learn Developers. (2024). *scikit-learn User Guide*.
<https://scikit-learn.org>

APPENDIX

Dataset View:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	track_id	artists	album_nar	track_nam	popularity	duration_m	explicit	danceabili	energy	key	loudness	mode	speechine	acousticne	instrument	liveness	valence	tempo	time_signa	track_genr
2	55uOikwR	Gen Hoshi	Comedy	Comedy	73	3.84	FALSE	0.676	0.461	1	-6.746	0	0.143	0.0322	1.01E-06	0.358	0.715	87.917	4	acoustic
3	4qPNDBW	Ben Wood	Ghost (Acc	Ghost - Ac	55	2.49	FALSE	0.42	0.166	1	-17.235	1	0.0763	0.924	5.56E-06	0.101	0.267	77.489	4	acoustic
4	1iJBS7s7j	Ingrid Mich	To Begin A	To Begin A	57	3.51	FALSE	0.438	0.359	0	-9.734	1	0.0557	0.21	0	0.117	0.12	76.332	4	acoustic
5	6lfqx3CG4	Kina Granr	Crazy Rich	Can't Help	71	3.37	FALSE	0.266	0.0596	0	-18.515	1	0.0363	0.905	7.07E-05	0.132	0.143	181.74	3	acoustic
6	5vjLSffimll	Chord Ove	Hold On	Hold On	82	3.31	FALSE	0.618	0.443	2	-9.681	1	0.0526	0.469	0	0.0829	0.167	119.949	4	acoustic
7	01MVOI9K	Tyrone We	Days I Will	Days I Will	58	3.57	FALSE	0.688	0.481	6	-8.807	1	0.105	0.289	0	0.189	0.666	98.017	4	acoustic
8	6Vc5wAmi	A Great Big	Is There Ar	Say Someti	74	3.82	FALSE	0.407	0.147	2	-8.822	1	0.0355	0.857	2.89E-06	0.0913	0.0765	141.284	3	acoustic
9	1EzrEOXm	Jason Mraz	We Sing. W	I'm Yours	80	4.05	FALSE	0.703	0.444	11	-9.331	1	0.0417	0.559	0	0.0973	0.712	150.96	4	acoustic
10	0lktbUcnA	Jason Mraz	We Sing. W	Lucky	74	3.16	FALSE	0.625	0.414	0	-8.7	1	0.0369	0.294	0	0.151	0.669	130.088	4	acoustic
11	7k9GuJYL	Ross Copp	Hunger	Hunger	56	3.43	FALSE	0.442	0.632	1	-6.77	1	0.0295	0.426	0.00419	0.0735	0.196	78.899	4	acoustic
12	4mzP5mH	Zack Tabur	Episode	Give Me Yc	74	4.08	FALSE	0.627	0.363	8	-8.127	1	0.0291	0.279	0	0.0928	0.301	99.905	4	acoustic
13	5ivF4eQB	Jason Mraz	Love Is a F	I Won't Giv	69	4	FALSE	0.483	0.303	4	-10.058	1	0.0429	0.694	0	0.115	0.139	133.406	3	acoustic
14	4ptDJbJl	Dan Berk	Solo	Solo	52	3.31	FALSE	0.489	0.314	7	-9.245	0	0.0331	0.749	0	0.113	0.607	124.234	4	acoustic
15	0X9MxHR1	Anna Ham	Bad Liar	Bad Liar	62	4.14	FALSE	0.691	0.234	3	-6.441	1	0.0285	0.777	0	0.12	0.209	87.103	4	acoustic
16	4LbWtBkN	Chord Ove	Hold On (R	Hold On - F	56	3.14	FALSE	0.755	0.78	2	-6.084	1	0.0327	0.124	2.83E-05	0.121	0.387	120.004	4	acoustic
17	1KHdq8NK	Landon Pig	The Boy W	Falling in L	58	4.08	FALSE	0.489	0.561	4	-7.933	1	0.0274	0.2	4.56E-05	0.179	0.238	83.457	3	acoustic
18	6xKeQgzfj	Andrew Fo	ily (i love	yc ily (i love	56	2.16	FALSE	0.706	0.112	2	-18.098	1	0.0391	0.827	4.03E-06	0.125	0.414	110.154	4	acoustic
19	4YoOigmcc	Andrew Fo	At My Wors	At My Wors	54	2.83	FALSE	0.795	0.0841	10	-18.09	0	0.0461	0.742	1.17E-05	0.0853	0.609	91.803	4	acoustic
20	2qLM6TuE	Jason Mraz	We Sing. W	Lucky	68	3.16	FALSE	0.625	0.414	0	-8.7	1	0.0369	0.294	0	0.151	0.669	130.088	4	acoustic
21	6CgNoAbF	Boyce Ave	Cover Sest	Photograp	67	4.34	FALSE	0.717	0.32	3	-8.393	1	0.0283	0.83	0	0.107	0.322	107.946	4	acoustic
22	3500XQec	Jason Mraz	We Sing. W	I'm Yours	75	4.05	FALSE	0.703	0.444	11	-9.331	1	0.0417	0.559	0	0.0973	0.712	150.96	4	acoustic
23	210JCw2L	Boyce Ave	Cover Sest	Demons	63	2.9	FALSE	0.678	0.351	0	-8.654	1	0.0266	0.747	0	0.355	0.569	90.032	4	acoustic
24	5TVe3pk05	A Great Big	Is There Ar	Say Someti	70	3.82	FALSE	0.407	0.147	2	-8.822	1	0.0355	0.857	2.89E-06	0.0913	0.0765	141.284	3	acoustic
25	0BUuuEvN	Jason Mraz	Coffee Mo	93 Million	0	3.61	FALSE	0.572	0.454	3	-10.286	1	0.0258	0.477	1.37E-05	0.0974	0.515	140.182	4	acoustic
26	3Hn3Lfh	Q Jason Mraz	Human - B	Unlonely	0	3.85	FALSE	0.796	0.667	5	-4.831	0	0.0392	0.381	0	0.221	0.754	97.988	4	acoustic
27	6D33wCKz	Jason Mraz	Mellow Ad	Bella Luna	1	5.04	FALSE	0.755	0.454	9	-9.609	0	0.0352	0.757	0	0.236	0.33	120.06	4	acoustic

Fig 14: Portion of actual dataset

Model Results:

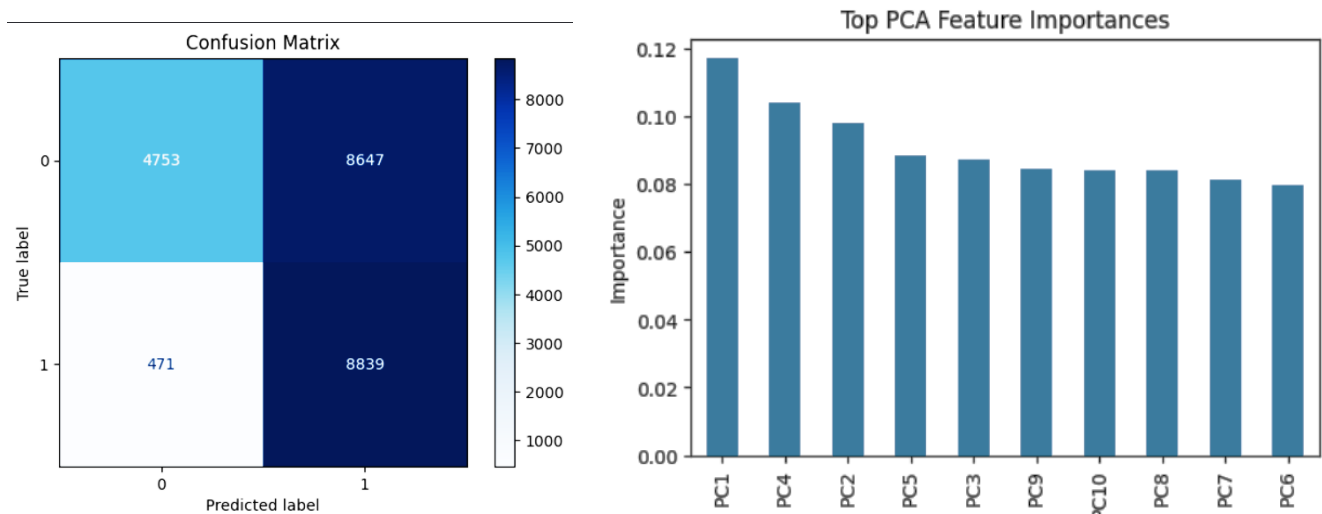


Fig 15: (i) Correlation matrix to show how accurate the model predicts
(ii) Bar chart to show how important the principle components are

SAMPLE DATA

```
new_song = pd.DataFrame([{
    'duration_ms': 205000, 'explicit': 0, 'danceability': 0.86, 'energy': 0.92,
    'key': 7, 'loudness': -4.3, 'mode': 1, 'speechiness': 0.05,
    'acousticness': 0.08, 'instrumentalness': 0.0002, 'liveness': 0.12,
    'valence': 0.81, 'tempo': 124, 'time_signature': 4,
    'duration_min': 205000/60000, 'energy_valence_ratio': 0.92/(0.81+0.01),
    'acousticness_inverse': 1-0.08, 'dance_energy_mix': (0.86+0.92)/2,
    'tempo_norm': (124 - df['tempo'].mean()) / df['tempo'].std()
}])
```

Fig 16: Sample input data to see whether the trained model gives accurate result

LIBRARIES & MODULES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, f1_score, roc_auc_score, classification_report, confusion_matrix, ConfusionMatrixDisplay
from sklearn.cluster import KMeans
from imblearn.over_sampling import SMOTE
import joblib
from sklearn.metrics import RocCurveDisplay
from sklearn.metrics import PrecisionRecallDisplay
import scipy.stats as stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.multivariate.manova import MANOVA
```

Fig 17: All libraries and modules implemented across entire project